

MONIKONGKONA HAZARIKA

Principal Software Engineer | Data Platforms, Orchestration, and Machine Learning

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PROFESSIONAL SUMMARY

Principal Software Engineer with 15+ years of experience designing and delivering distributed systems, scalable microservices, cloud-native platforms, and production data pipelines. Proven technical leader in data ingestion, search platforms, lakehouse architecture, workflow orchestration, and applied machine learning. Track record of leading cross-functional engineering teams, driving architectural decisions, and delivering mission-critical systems with reliability, scalability, and performance. Seeking Senior Principal, Architect, or technical leadership roles.

CORE COMPETENCIES

- Data Engineering: MongoDB, Elasticsearch, CDC, ETL/ELT, Data Lakehouse, S3, Parquet, Iceberg, Spark SQL, ClickHouse, Trino, Athena
- Orchestration and Platforms: Temporal, Kubernetes, Docker, Kafka, RabbitMQ, Debezium, Microservices, Spring Boot
- Machine Learning: Regression, Classification, Clustering, Neural Networks, Deep Learning, A/B Testing, H2O AutoML, TensorFlow, Keras
- Cloud and Infrastructure: AWS, Oracle Cloud (OCI), Terraform, CI/CD (Jenkins, GitHub Actions), Argo
- Observability and Reliability: Prometheus, Grafana, OpenTelemetry, ELK Stack, SLA/SLO, Incident Management, RCA
- Leadership: Technical Leadership, Team Mentorship, Architecture, Stakeholder Management, Platform Strategy

PROFESSIONAL EXPERIENCE

Swimlane

Principal Software Engineer | April 2024 - Present | Bangalore, India

- Lead Data and Orchestration engineering initiatives, reporting to VP Engineering and driving platform roadmap, architecture, and delivery.
- Designed and operated a high-availability data ingestion platform from MongoDB to Elasticsearch using CDC, RabbitMQ, and Kubernetes with KEDA autoscaling, achieving 99.9% uptime and SLA adherence.
- Improved search performance 2x and reduced query latency by 50% through pipeline and indexing optimizations.
- Built observability-first solutions with Prometheus, OpenTelemetry, and Grafana dashboards for proactive alerting, RCA, and performance tuning.
- Re-architected pipelines and storage strategies to reduce infrastructure costs while maintaining scalability.
- Developed and maintained Kubernetes CronJobs and microservices for ingestion pipelines with governance and data quality monitoring.
- Built a self-hosted Elasticsearch ingestion pipeline for audit logs, including cluster sizing, monitoring, and SRE metrics.
- Led POC to migrate MongoDB Records collection to Amazon S3 as Parquet files and Apache Iceberg tables, using Spark SQL and ClickHouse for analytics and reporting; exploring Trino and Starburst.
- Building AI agent solutions in the security domain.

Zuora

Principal Software Engineer | September 2021 - February 2024 | Bangalore, India

- Led platform modernization by designing microservices with Temporal workflows, Spring Boot, and MongoDB for fault tolerance and scalability.
- Delivered 5x performance improvement by transforming PL/SQL logic into Java using Temporal Java SDKs; adopted MongoDB for RevPro product workloads.
- Reduced job runtime from 12 hours to 4 hours (66% latency improvement) and implemented Jenkins pipelines for end-to-end regression automation.
- Built a Kafka-based CDC archival pipeline, improving storage efficiency 3x while reducing infrastructure spend.
- Mentored a team of 10 engineers on data pipeline design, CI/CD, and observability practices.
- Owned stakeholder communications and platform KPIs for performance, adoption, and business alignment.

Oracle

Principal Software Engineer | December 2014 - September 2021 | Bangalore, India

- Contributed to migration of Oracle Responsys SaaS Cloud Marketing platform from on-premise data centers to Oracle Cloud Infrastructure (OCI).
- Designed and implemented platform components supporting massive scale, including 9.7B+ marketing messages during Cyber Weekend and 1.7B+ messages in a single day.
- Built orchestration engines and platform services for Oracle Management Cloud (OMC) with focus on reliability, scalability, and performance.
- Developed ML models including an omnichannel prediction model using neural networks and multivariate testing POCs with statistical significance.
- Explored push notification title prediction using Neural Networks, CountVectorizer, Word2Vec, GloVe, Random Forest, and XGBoost.
- Applied multi-threading, transaction management, and performance debugging expertise; supported production systems, RCA, and incident resolution.
- Contributed to intelligent skew optimization and large-scale data pipelines for Oracle Cloud services.

British Telecom

Consultant | June 2013 - December 2014

- Developed multiple on-premise Java projects from scratch with strong design and development contributions.
- Delivered telecom product design and J2EE development including Spring-JDBC, web services, and XML parsing.

Ericsson

Software Developer | July 2010 - June 2013

- Worked on Java-based BMC Remedy projects for global customers including AT&T, Endesa Spain, and HPSC.
- Key contributor in Ericsson IPTV development for Cellcom Israel.

CAREER HIGHLIGHTS

- Anchor Award, Swimlane Q2 2025.
- 3rd prize, Zuora Z-Embark innovation program, March 2023.
- Best Employee Award, British Telecom, 2013-2014.
- Consecutive 3-year best performer and MVP award, Oracle Responsys team.
- Winner, Oracle Hackathon 2017 (Intelligent Multivariate Testing Email with statistical significance, later productized).
- Finalist, Oracle Hackathon 2018 (Goal-based Marketing).

- Winner, Oracle Hackathon 2019 (Predicting the best marketing program workflow).
- State Rank Holder in 10+2, Assam; Chemistry Olympiad winner; National Merit Scholarship, MHRD.

TECHNICAL SKILLS

- Programming: Python, Java, .NET, PL/SQL, Shell
- Frameworks: Spring (DI, AOP, Transaction Management), Spring Boot, Spring JDBC, JDB, Maven
- Databases: MongoDB, PostgreSQL, Oracle, Elasticsearch, Redis
- Big Data and Analytics: Spark SQL, Trino, ClickHouse, Amazon Athena, Apache Calcite, Hive, Parquet, Iceberg
- Messaging and Streaming: Kafka, RabbitMQ, Debezium
- ML and Data Science: scikit-learn, NumPy, pandas, Matplotlib, H2O AutoML, Keras, TensorFlow, Jupyter
- DevOps and Platform: Kubernetes, Docker, Terraform, Jenkins, GitHub Actions, Argo, Temporal.io
- Observability: Prometheus, Grafana, OpenTelemetry, ELK Stack, Structured Logging, Zabbix
- Tools: IntelliJ, Rider, Eclipse, SQL Developer, Postman, JMeter, VisualVM, Git, SVN
- Languages: English, Hindi, Assamese, Bengali

EDUCATION

- Diploma in AI-ML (WILP), BITS Pilani, 2019-2020
- B.Tech. Computer Science, NIT Silchar, 2006-2010

CERTIFICATIONS

- Machine Learning, Coursera (Stanford University), 2020 - Completed with 95%
- Object-Oriented Design, 2021
- Microservices Foundation, 2021
- Oracle Database, 2021
- Oracle Cloud Infrastructure Foundations, 2021